

PERCENT COMPOSITION

Instructions: Show all work step by step. Write neatly; use a pencil for all calculation, and a ruler for any underlining that may be required. You must also be mindful of the number of sig. figs in your final answer.

1. What is the percent composition of an ore, which contains 1.85 grams of aluminum & 1.65 grams of oxygen?
2. A compound of silver contains 4.35 grams of that metal combined with 0.65 grams of sulfur. What is the percent composition of that compound?
3. A certain sample of gas weighing 22.8 grams contains 6.22 grams of carbon and an unweighted quantity of oxygen. What is the percent composition of the gas?
4. A compound consisting of aluminum & chlorine weighs 17.82 grams. The aluminum in the compound weighs 3.60 grams. What is the weight of the chlorine in the compound? What is the percent composition of the compound?
5. An organic compound consisting of carbon, hydrogen, and oxygen only weighs 13.669 grams. Analysis shows that the compound contains 0.547 grams and 8.707 grams of the last two elements respectively. What is the percentage composition of the compound?
6. A sample of zinc sulfide having a mass of 3.04 grams was analyzed and found to contain 1.00 grams of sulfur and the remainder zinc. If a prospector discovered 1000 Kg of a mineral containing pure zinc sulfide, what would be the maximum amount of zinc that could be recovered from this rock?